

A sharp center consequence for almost band preservers on $C_0(K)$

Partial result packet for arXiv:1610.02557

Run: fa_banach_001

Source and target question

The source paper is Timur Oikhberg and Pedro Tradacete, *Almost band preservers*, arXiv:1610.02557 [1]. It defines a linear map $T : X \rightarrow X$ on a Banach lattice to be ε -band preserving if

$$\sup\{\| |Tx| \wedge y \| : y \geq 0, y \perp x\} \leq \varepsilon \|x\| \quad (x \in X).$$

It also defines $T \in \varepsilon - \mathcal{Z}(X)$ when there is $\lambda > 0$ such that for every $x \in \mathbf{B}(X)$ there is $z \in X$ with $\|z\| \leq \varepsilon$ and

$$|Tx| \leq \lambda |x| + z.$$

Equivalently, as stated in the source paper, $T \in \varepsilon - \mathcal{Z}(X)$ iff there is $\lambda \geq 0$ such that

$$\sup_{x \in \mathbf{B}(X)} \|(|Tx| - \lambda |x|)_+\| \leq \varepsilon.$$

The relevant source passage is:

“In general, if $T \in \varepsilon - \mathcal{Z}(X)$, then T is ε -BP. We do not know whether the converse implication holds in general.”

With properties, we introduce:

Definition 4.2. *An operator $T : X \rightarrow X$ is in the ε -center (in short $T \in \varepsilon - \mathcal{Z}(X)$) if there exists $\lambda > 0$ such that for every $x \in \mathbf{B}(X)$, there is $z \in X$ with $\|z\| \leq \varepsilon$ such that*

$$|Tx| \leq \lambda |x| + z.$$

Note that $T \in 0 - \mathcal{Z}(X) = \mathcal{Z}(X)$ if and only if T is BP ([1]). Moreover, if T is BP, and S is arbitrary, then $T + S \in \|S\| - \mathcal{Z}(X)$. In general, if $T \in \varepsilon - \mathcal{Z}(X)$, then T is ε -BP. We do not know whether the converse implication holds in general. However, if $T \in ABP(\varepsilon)$, then $T \in \varepsilon - \mathcal{Z}(X)$. In Section 5 we will provide conditions for which every ε -BP operator is in $ABP(4\varepsilon)$, hence it also belongs to $4\varepsilon - \mathcal{Z}(X)$.

Note that $T \in \varepsilon - \mathcal{Z}(X)$ if and only if there is $\lambda \geq 0$ such that for every $x \in \mathbf{B}(X)$,

$$\|(|Tx| - \lambda |x|)_+\| \leq \varepsilon.$$

For $T \in \varepsilon - \mathcal{Z}(X)$, we define

$$\rho_\varepsilon(T) = \inf\{\lambda \geq 0 : \sup_{x \in \mathbf{B}(X)} \|(|Tx| - \lambda |x|)_+\| \leq \varepsilon\}.$$

The source later proves that on compact $C(K)$, every bounded ε -BP operator is within 2ε of a band-preserving multiplication operator, and therefore belongs to $2\varepsilon - \mathcal{Z}(C(K))$ [1, Theorem 6.2 and following remark]. The result below records the sharper center consequence suggested by the measure estimate inside that proof, and extends it to locally compact $C_0(K)$ spaces.

Result

Theorem 1 (Sharp $C_0(K)$ center consequence). *Let K be a locally compact Hausdorff space and let $\varepsilon \geq 0$. If*

$$T : C_0(K) \longrightarrow C_0(K)$$

is a linear ε -band preserving map, then $T \in \varepsilon - \mathcal{Z}(C_0(K))$. More precisely, for every $f \in C_0(K)$ with $\|f\|_\infty \leq 1$,

$$\|(|Tf| - \|T\| |f|)_+\|_\infty \leq \varepsilon.$$

Corollary 1 (AL-spaces). *Let X be an AL-space and let $\varepsilon \geq 0$. Every linear ε -band preserving map $T : X \rightarrow X$ belongs to $\varepsilon - \mathcal{Z}(X)$.*

Remark 1 (Sharpness). *The constant ε in Theorem 1 is optimal, even for ℓ_∞^2 . For $\varepsilon > 0$, define*

$$T(a, b) = (\varepsilon b, 0), \quad (a, b) \in \ell_\infty^2.$$

Then T is ε -BP. However, for $x = (0, 1)$ and every $\lambda \geq 0$,

$$(|Tx| - \lambda|x|)_+ = (\varepsilon, 0),$$

so $T \notin c - \mathcal{Z}(\ell_\infty^2)$ for any $c < \varepsilon$.

Proof intuition

For $C_0(K)$, boundedness is automatic by the source paper. At each point $t \in K$, the scalar functional $f \mapsto Tf(t)$ is represented by a regular Radon measure μ_t . Split off the atom at t :

$$\mu_t = a_t \delta_t + \nu_t, \quad \nu_t(\{t\}) = 0.$$

The ε -BP condition says that a function vanishing at t cannot produce more than ε at t after applying T . This controls the off-atom part: $\|\nu_t\| \leq \varepsilon$. Hence

$$|Tf(t)| \leq |a_t| |f(t)| + \varepsilon \|f\|_\infty,$$

and $|a_t| \leq \|T\|$. That pointwise inequality is exactly the ε -center inequality.

Proof

We first record the locally compact version of the scalar vanishing criterion used in the compact $C(K)$ proof of [1, Lemma 6.1].

Lemma 1. *Let $T : C_0(K) \rightarrow C_0(K)$ be linear and ε -BP. Then T is bounded, and for every $t \in K$ and every $f \in C_0(K)$ with $f(t) = 0$,*

$$|Tf(t)| \leq \varepsilon \|f\|_\infty.$$

Proof. The boundedness is [1, Theorem 3.13], applied to the scalar space $\mathbb{R}$.

If f vanishes on an open neighborhood V of t , the desired estimate is exactly [1, Lemma 3.9], again in the scalar case.

Now suppose only that $f(t) = 0$. Fix $\delta > 0$. By continuity, choose an open neighborhood U of t such that $|f(s)| < \delta$ on U . Since K is locally compact Hausdorff, choose $h \in C_c(K)$ with $0 \leq h \leq 1$, $h = 1$ on a neighborhood of t , and $\text{supp } h \subset U$. Put $g = (1 - h)f$. Then g vanishes on a neighborhood of t , $\|g\|_\infty \leq \|f\|_\infty$, and $\|f - g\|_\infty = \|hf\|_\infty \leq \delta$. Therefore

$$|Tg(t)| \leq \varepsilon \|f\|_\infty, \quad |T(f - g)(t)| \leq \|\delta_t T\| \delta.$$

Letting $\delta \downarrow 0$ gives the assertion. $\square$

Proof of Theorem 1. By Lemma 1, T is bounded. Fix $t \in K$. The bounded functional $f \mapsto Tf(t)$ on $C_0(K)$ is represented by a regular finite signed (or complex) Radon measure μ_t :

$$Tf(t) = \int_K f d\mu_t, \quad f \in C_0(K).$$

Let $a_t = \mu_t(\{t\})$ and $\nu_t = \mu_t - a_t \delta_t$. Then $\nu_t(\{t\}) = 0$.

We claim that $\|\nu_t\| \leq \varepsilon$. Let $f \in C_0(K)$ with $\|f\|_\infty \leq 1$, and let $\eta > 0$. By regularity of the total variation measure $|\nu_t|$, and since $|\nu_t|(\{t\}) = 0$, choose an open neighborhood U of t such that $|\nu_t|(U) < \eta$. Choose $h \in C_c(K)$ with $0 \leq h \leq 1$, $h(t) = 1$, and $\text{supp } h \subset U$. Then

$$|\nu_t(fh)| \leq |\nu_t|(U) < \eta.$$

Also $f(1 - h)$ vanishes at t , so Lemma 1 gives

$$|\mu_t(f(1 - h))| = |T(f(1 - h))(t)| \leq \varepsilon.$$

Because $\delta_t(f(1 - h)) = 0$, this is the same as $|\nu_t(f(1 - h))| \leq \varepsilon$. Hence

$$|\nu_t(f)| \leq \varepsilon + \eta.$$

Letting $\eta \downarrow 0$ and taking the supremum over the unit ball of $C_0(K)$ proves $\|\nu_t\| \leq \varepsilon$.

For $\|f\|_\infty \leq 1$, we therefore have pointwise

$$|Tf(t)| = |a_t f(t) + \nu_t(f)| \leq |a_t| |f(t)| + \varepsilon.$$

Moreover $|a_t| \leq \|\mu_t\| = \|\delta_t T\| \leq \|T\|$. Thus

$$|Tf(t)| \leq \|T\| |f(t)| + \varepsilon \quad (t \in K, \|f\|_\infty \leq 1),$$

which is equivalent to

$$\|(|Tf| - \|T\| |f|)_+\|_\infty \leq \varepsilon.$$

This is precisely $T \in \varepsilon - \mathcal{Z}(C_0(K))$. $\square$

Proof of Corollary 1. For $\varepsilon > 0$, an ε -BP linear map on an order continuous Banach lattice is bounded by [1, Corollary 3.8]; AL-spaces are order continuous. For $\varepsilon = 0$, boundedness of band-preserving maps is the Abramovich–Veksler–Koldunov theorem quoted in the source introduction. Thus $T \in B(X)$.

Since X is order continuous, [1, Proposition 2.11] implies that $T^* : X^* \rightarrow X^*$ is ε -BP. The dual of an AL-space is an AM-space with unit, hence is lattice isometric to $C(L)$ for a compact Hausdorff space L by Kakutani's representation theorem. Applying Theorem 1 to this compact $C(L)$ gives

$$T^* \in \varepsilon - \mathcal{Z}(X^*).$$

Finally, [1, Proposition 4.3] says that $S \in \varepsilon - \mathcal{Z}(Y)$ iff $S^* \in \varepsilon - \mathcal{Z}(Y^*)$. Applying the reverse direction to T gives $T \in \varepsilon - \mathcal{Z}(X)$. $\square$

Verification notes and limitations

The proof is a locally compact adaptation of the measure estimate already present in [1, Theorem 6.2]. The source paper used this estimate to approximate T by the multiplication operator $f \mapsto (T1)f$, losing a second ε . For the center conclusion that approximation step is unnecessary: the off-atom estimate alone gives the sharp ε -bound.

This packet does not solve the source paper's general question for arbitrary Banach lattices. It gives the sharp positive answer for $C_0(K)$ spaces and, by duality, for AL-spaces. Local run-index searches for the arXiv id and core phrases found no duplicate packet. A bounded web search for exact title and keyword combinations found the source arXiv record and no later exact resolution of this sharpened statement.

References

- [1] T. Oikhberg and P. Tradacete, *Almost band preservers*, arXiv:1610.02557, 2016. <https://arxiv.org/abs/1610.02557>